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MAT: Mixing Attention Transfer from Multiple Transformers for Medical Tasks

Zi-Hao Bo, Yuchen Guo, Xiangru Chen, Jing Xie, Lishan Ye, and Feng Xu

***Abstract*— Transformer has been widely used for image analysis tasks, but in medicine, it suffers from limited data availability. To overcome this challenge, we propose a novel approach specially designed for transformers to transfer knowledge from multiple sources to target medical tasks with limited data, named Mixing Attention Transfer (MAT). MAT aims to harness and merge knowledge from multiple source transformers at the token and layer level to improve the performance of target medical tasks. The core component of MAT is the Mixing Attention layer, which encompasses: 1. token-level Routing and Fusion modules that allocate input images to adequate source modules; 2. sequence-level Aligned-Attention module that adaptively aligns outputs produced by different source modules. To the best of our knowledge, this is the first multi-source transfer learning approach specifically designed for transformers. Through extensive evaluations, we demonstrate the effectiveness of MAT on three common medical scenarios: noisy-labeled, class-imbalanced, and fine-grained tasks.**

***Index Terms*— multi-source transfer learning, transformer model fusion, medical image analysis**

I. Introduction

MEDICAL image analysis using deep learning has become increasingly popular in recent years [1], [2]. However, medical image datasets are typically small, especially for rare diseases [3]. Moreover, strict regulations for data collection and sharing further impede the feasibility of augmenting dataset sizes with multiple sources [4]. Consequently, the effective utilization of limited data is very important to construct robust deep-learning models for medical tasks.

The transformer architecture [5] has emerged as a widely adopted network architecture in various domains, including medical image analysis [6], [7]. For large-scale models, the transformer has demonstrated superior performance over convolutional networks [8], [9]. However, transformers often require a larger amount of training data to achieve their full potential [10]. Therefore, the utilization of transfer learning becomes crucial for using transformers in the medical domain [6]. In particular, knowledge transfer from multiple source tasks may significantly enhance the performance with limited data.

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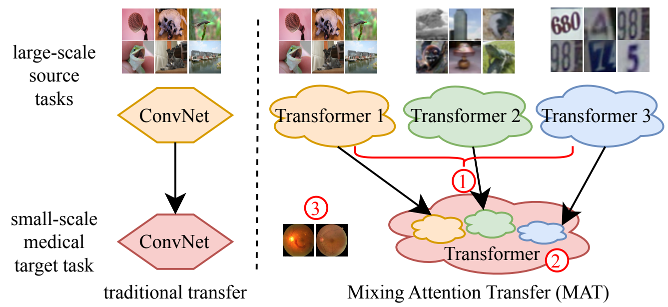

Fig. 1. Mixing Attention Transfer (MAT) learning - a solution for transferring knowledge (1) from multiple source tasks (2) for transformer models (3) on small-scale medical tasks.

Transfer learning from multiple source tasks is a persistent challenge in natural image processing. There exist three primary approaches aimed at addressing this problem: ensemble-based, selection-based, and fusion-based methods. Ensemble-based methods [11]–[14] are widely used to make use of multiple models. However, these methods only combine the final output of the source models, thereby potentially under-utilizing the knowledge embedded in the intermediate features of the source models. Selection-based approaches [15]–[20] employ sophisticated methods to measure transferability and identify the best model or sub-modules from the source models to transfer to the target. However, accurately quantifying transferability poses a challenge, and the hard selection of the best model may discard other source models that could potentially contain valuable information. Recently, fusion-based methods [21]–[23] have been introduced to combine the parameters or features of multiple source models. However, these approaches are often limited to fusing models trained on the same task. In general, all these techniques are not specifically tailored for transformer models, and may not fully exploit the abundant information present in the token- and layer-level features of transformer models when applied to small-scale medical tasks.

We introduce Mixing Attention Transfer (MAT) learning (see in Figure 1) as a solution specifically designed for transformer models to transfer knowledge from multiple source

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tasks to small-scale medical tasks. Due to limited data for medical tasks, utilizing multiple source tasks is crucial to the final performance. MAT incorporates the Mixing Attention layer which fully fuses the knowledge of multiple source transformers at the token and layer level. Firstly, as each source model may possess distinct strengths and weaknesses, we employ a Routing module to allocate each input token to the appropriate sources, and a Fusion module to aggregate their output tokens based on the appropriateness. Then, to avoid the misalignment of output tokens from different source modules that could negatively impact training, we introduce the Aligned-Attention module to dynamically align them at the sequence level. Lastly, we incorporated the Mixing Attention into each transformer layer to integrate intricate information at the layer level. Compared to existing methods, MAT is the first to enable multiple-source knowledge fusion and transfer at the token- and layer-level. It extends the multi-source ability of vanilla MoE from only the FFN layer to the entire transformer model. This extension is made feasible by the proposed Aligned-Attention module, which enables multi-source routing for QKV-MLP without harming the performance. In this manner, MAT effectively leverages the unique strengths of each source model.

Our contributions can be summarized as follows:

- We propose Mixing Attention Transfer (MAT) to enable transformer models to achieve transfer learning from multiple source tasks. MAT leverages the unique strengths of each source, enabling robust and powerful transformer model training for small-scale target medical tasks.
- We propose the Mixing Attention layer for MAT, which integrates Routing and Fusion modules to allocate input images to different source modules and aggregate them at the token level. Moreover, it incorporates Aligned-Attention modules to dynamically align the output token features at the sequence level.
- We demonstrate the effectiveness of MAT and its superiority over the state-of-the-art methods on three common medical tasks: noisy-labeled, class-imbalanced, and fine-grained tasks with small-scale datasets.

II. Related work

Transfer learning aims to improve the performance of a target task by leveraging knowledge from a source task [24]. Traditional transfer learning approaches are often transferred from a single source model. However, in small-scale medical tasks, transferring knowledge from multiple source tasks can be advantageous, especially for transformer models.

Model fusion. Model fusion is a straightforward approach to combine multiple models into a single one. For instance, OT-Fusion [25] merges parameters from source models using optimal transport. Git Re-Basin [26] introduces three matching methods to find the optimal permutation. CLAFusion [27] utilizes cross-layer alignment to fuse heterogeneous neural networks. However, model fusion approaches are limited to merging models trained on the same task. Although recent work ZipIt is proposed to fuse models trained on different tasks [28], its performance cannot overtake ensemble methods [11]. Additionally, these approaches are primarily designed for small convolutional models and are still in nascent stages for real-world applications.

Transfer learning from Model Zoo. In recent years, several approaches have been proposed for transferring knowledge from multiple models pretrained on different source tasks. One popular solution involves evaluating the transferability of models or sub-models and selecting the best one for the target task [15], [16], [18], [20]. These approaches do not require fine-tuning all the source models. However, selecting the best model may not fully leverage and fuse knowledge from multiple source tasks. Model soups [23] demonstrates the effectiveness of parameter averaging when fusing multiple models. However, simple averaging is only effective when applied to the same source tasks. Alternatively, other approaches involve aggregating features or parameters using sophisticated methods. For example, Knowledge-Flow [21] incorporates knowledge from source models by merging their intermediate layer features. Zoo-Tuning [22] uses adaptive aggregation to fuse parameters. These methods perform well for convolutional models on natural image analysis. However, they are not specifically designed for transformer models. Hub-Pathway [14] proposes a routing method to fuse models with any structure. However, its routing is on the model level, similar to ensembling methods [11], and is unable to fully exploit complex knowledge within models. Consequently, further exploration is required for multi-source transfer learning of transformers for small-scale medical tasks.

Mixture of experts. Mixture of experts (MoE) [29] is a popular approach for constructing large sparsely activated transformer models, though it was not originally proposed for solving multi-source transfer learning problems. In the MoE model, parameters are partitioned into multiple "experts" that specialize in different aspects of the task [30], [31]. MoE approaches have been widely used in natural language processing [32], [33], computer vision [34], [35], and multi-modal learning [36]. In medical image analysis, MoE was also adopted to multi-task and multi-modal learning [37]–[39]. However, the vanilla MoE was originally proposed as a single sparse model trained on a single task or multiple tasks simultaneously [40], rather than for transfer learning purposes. In addition, some parameters, such as those in the input embedding and self-attention layer, are not partitioned into experts [41]. As a result, it cannot be directly applied to multi-source transfer learning. To address this, we propose MAT, which allows us to leverage the entire transformer model from multi-source models.

III. Method of Mixing Attention Transfer

In this section, we present the problem setting for MAT, the overall architecture of MAT, and the detailed design of the Mixing Attention layer within MAT.

A. Multi-Source Transfer for Medical Tasks

In medical tasks, data availability is often limited. Thus, it is crucial to fuse and reuse knowledge from the source task to train robust transformer models for medical tasks. Unlike

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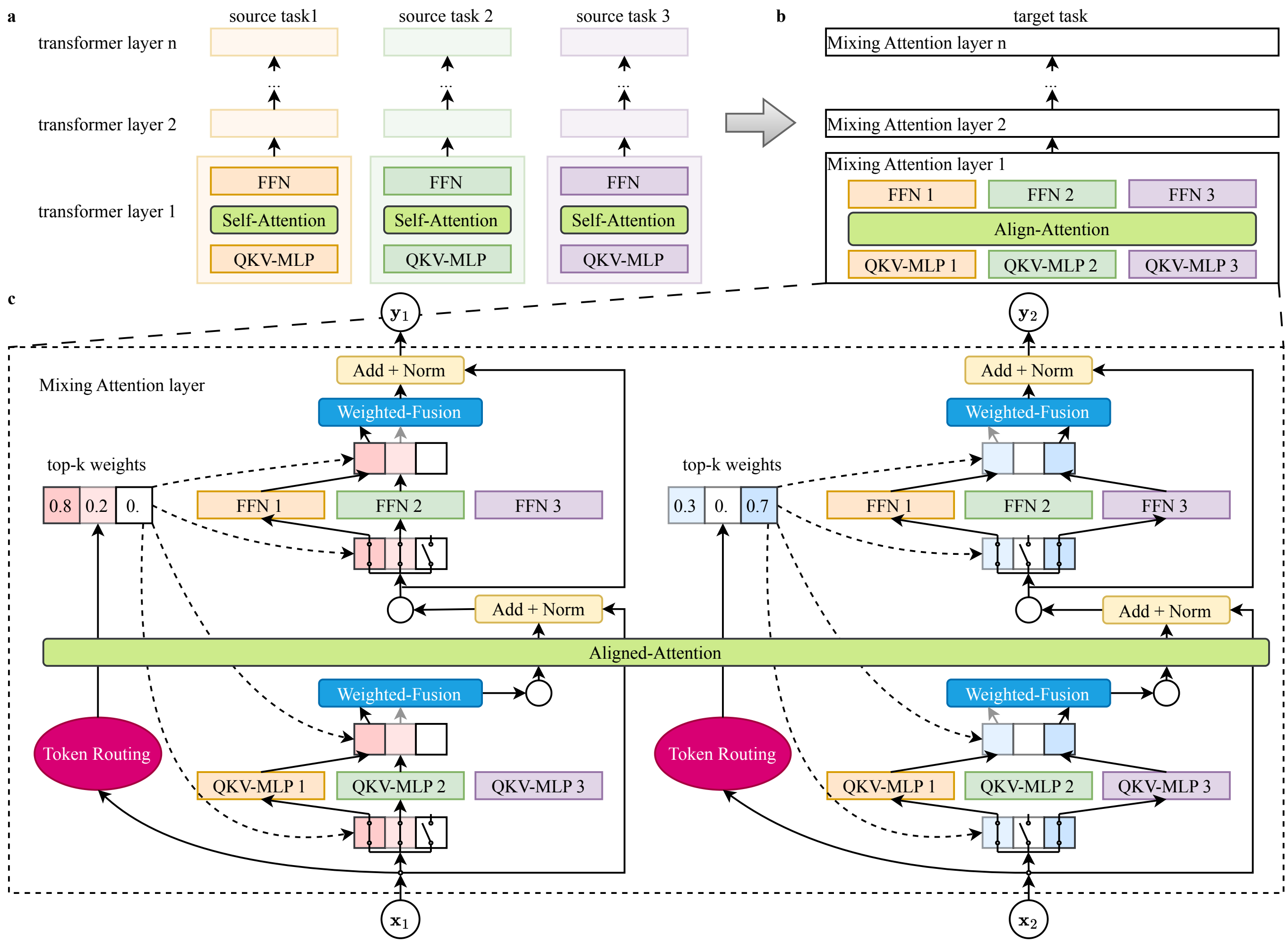

Fig. 2. Overview of Mixing Attention Transfer (MAT). **a**. MAT leverages multiple pretrained source transformer models trained on different source tasks. **b**. MAT incorporates Mixing Attention layers to merge the transformer layers from the source models. **c**. Each Mixing Attention layer comprises token-level Routing, Fusion, and sequence-level Aligned-Attention modules. The Token-Routing module analyzes each input token and assigns it to top-k best suitable source modules. The Fusion module aggregates the above top-k output tokens. The Aligned-Attention module aligns token features processed by different source modules at the sequence level. The routing is done for both QKV-MLP and FFN. $\mathbf{x_1}$, $\mathbf{x_2}$, ...: token features from an input sequence; $\mathbf{y_1}$, $\mathbf{y_2}$, ...: corresponding output tokens.

existing methods that only consider merging the final output or model weights statically, MAT is specifically designed to dynamically leverage the structural knowledge stored within transformer models trained on different source tasks, token- and layer-wise.

Therefore, MAT jointly merge all of the source models at the layer-level, using Mixing Attention layers. Figure 2 shows the overall architecture of MAT. Assuming we have M transformer models $\{\mathcal{M}_{src}^{(i)}\}_{i=1}^{M}$ trained on M source tasks $\{\mathcal{T}_{src}^{(i)}\}_{i=1}^{M}$, our goal is to fuse them and build a target model $\mathcal{M}_{tgt} = \text{MAT}(\{\mathcal{M}_{src}^{(i)}\}_{i=1}^{M})$ that effectively leverage and exploit the knowledge contained within $\{\mathcal{T}_{src}^{(i)}\}_{i=1}^{M}$ to improve performance on the target task $\mathcal{T}_{tgt}$.

In our medical image analysis tasks, all the models are vision-transformer models featuring an input embedding layer, N transformer layers, and a classification head layer. The input image undergoes processing by the input embedding layer, resulting in T sequential tokens. Subsequently, the stacked transformer layers handle the input tokens, followed by the classification head for predicting the target label. For clarity, only the transformer layers are depicted in Figure 2. The transformer layer can be denoted as $\{\mathbf{y}_t\}_{t=1}^{T} = Trans(\{\mathbf{x}_t\}_{t=1}^{T})$, where $\mathbf{x}$ and $\mathbf{y}$ are the input and output tokens. As Figure 2a shows, we decompose each transformer layer into a QKV-MLP module, a Self-Attention module, and an FFN module. To transfer $\{\mathcal{M}_{src}^{(i)}\}_{i=1}^{M}$ to $\mathcal{M}_{tgt}$, we integrate N Mixing Attention layers, which consider the internal structure of transformer layers to fuse them at the layer level.

B. Mixing Attention Layer

To fully leverage the knowledge from all source models, we propose the concept of layer-wise Mixing Attention. Unlike previous methods that only merge models at the overall model level [11], [14], our approach fuses source models at both the layer and token levels, enabling a more fine-grained integration of intermediate representations.

Each Mixing Attention layer comprises three key components: the token-level Routing module, the token-level Fusion module, and the sequence-level Aligned-Attention module. The token-level Routing module is responsible for distributing

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input tokens to different source modules, ensuring that each source module receives relevant tokens based on its functionality. The token-level Fusion module combines the output tokens from all source modules, capturing collective knowledge. The Aligned-Attention module processes and aligns token features generated by different source modules at the sequence level, avoiding feature-space misalignment of output tokens that may affect model training.

Token-level Routing module. Token-level features are the unique units in transformer models that convey valuable information. Existing multi-source transfer learning methods [14], [20]–[22] do not cater specifically to transformers, without overlooking the exploitation of token-level features. Besides, the routing in vanilla MoE is only applied to the FFN layer. Therefore, within each Mixing Attention layer, we integrate a unified Routing module responsible for determining the most suitable source module, both for QKV-MLP and FFN, to process each input token. This token-level routing mechanism enables different sections of the input image to be processed by different source models. The routing process follows a top-k approach, allowing each input token to be processed by a maximum of k source models.

The Routing module takes an input token $\mathbf{x}_t$ and predicts its gating weights ω_t for all M source modules, which serve as a measure of the appropriateness that each source module can process $\mathbf{x}_t$. ω_t can vary for each input token. The token-level Routing is defined as:

$$\omega_t = \text{Softmax}(\text{TopK}(\text{Gate}(\mathbf{x}_t))) \tag{1}$$

where $\omega_t \in \mathbb{R}^M$ and $\text{Softmax}(\mathbf{z}) = \frac{e^{\mathbf{z}}}{\sum_{z \in \mathbf{z}} e^z}$. The function TopK sets non-top-k elements to $-\infty$ to zero their weights:

$$\text{TopK}(\mathbf{x})_j = \begin{cases} x_j & x_j \in \text{top-k elements of } \mathbf{x} \\ -\infty & \text{otherwise} \end{cases} \tag{2}$$

The Gate function projects the input token to M-dimensional logits, with a noise term added to prevent routing from collapsing to specific source modules:

$$\begin{aligned} \text{Gate}(\mathbf{x}_t) = & W_{gate} \cdot \mathbf{x}_t + \\ & n \cdot (\text{Softplus}(W_{noise} \cdot \mathbf{x}_t) + \epsilon) \end{aligned} \tag{3}$$

where $\text{Softplus}(\mathbf{z}) = \log(1+e^{\mathbf{z}})$. W_{gate} and W_{noise} represent the projection parameters for the gate and the noise terms, respectively. The term $n \sim N(0, 1)$ follows a standard normal distribution, and ϵ is the minimum noise added during training. In our experiments, we set ϵ to 0.01.

As shown in Figure 2c, we apply the Routing module to the QKV-MLP module and the FFN module separately. Both parts share the same routing weights within a layer. The Routing allows feature merging at the token level, surpassing previous methods that consider the overall feature.

Token-level Fusion module. Since each input token may be routed to several source modules, it is essential to combine their outputs. In contrast to ensemble-based approaches like Hub-Pathway [14], which only aggregate features at the final stage of the model, our method enables aggregation of output tokens at every layer and for each token individually. Routing and fusion at every layer and each token enable fine-grained and dynamic knowledge integration, improving transfer effectiveness.

To achieve this, we introduce a Token-level Fusion module after each Routing module, which aggregates the output tokens from the various source modules.

Specifically, it aggregates the outputs from the source modules ($\{\text{QKV-MLP}_i\}_{i=1}^M$ and $\{\text{FFN}_i\}_{i=1}^M$) by executing a weighted summation based on the corresponding ω_t values. Additionally, between the QKV-MLP and FFN, an Aligned-Attention module is introduced, which will be elaborated in the subsequent section:

$$\mathbf{q}_t, \mathbf{k}_t, \mathbf{v}_t = \sum_{i=1}^{M} \omega_t \cdot \text{QKV-MLP}_i(\mathbf{x}_t) \tag{4}$$

$$\mathbf{x}'_t = \text{Aligned-Attention}\left(\mathbf{q}_t, \{\mathbf{k}_i\}_{i=1}^T, \{\mathbf{v}_i\}_{i=1}^T\right) + \mathbf{x}_t \tag{5}$$

$$\mathbf{y}_t = \sum_{i=1}^{M} \omega_t \cdot \text{FFN}_i(\mathbf{x}'_t) + \mathbf{x}'_t \tag{6}$$

By utilizing this weighted Fusion module, the capabilities of each source model can be adaptively merged.

Sequence-level Aligned-Attention. The Routing and Fusion modules operate at the token level, which means that different input tokens may be processed by different source modules, leading to token features that follow different distributions. To address this issue, we need to align these token features at the sequence level. Some approaches, like Zoo-Tuning [22], employ extra parameters to transform models for alignment. However, this introduces additional computational complexity and cannot handle sequence-level features in transformer models. To address this, we propose the parameter-free Aligned-Attention module, which aligns the output tokens from different source modules at the sequence level.

How to design a self-attention mechanism to align tokens generated by different source modules? There are two intuitive options: 1. Restrict routing and fusion in the FFN, instead of the self-attention module, which prevents the fusion of self-attention knowledge from multi-source tasks. 2. Routing and fusion to the entire self-attention module, which would require all tokens in a sequence to follow the same routing weights to avoid incomplete self-attention operations. Mixture of experts [29] follows the first option and only routes tokens in the FFN layer. While this is suitable for single- or multi-task learning, it cannot fully leverage multiple QKV-MLP layers in multi-source transfer learning. Some approaches, such as model ensemble [11] and Hub-Pathway [14], adopt methods similar to the second option, where routing is performed for the entire model and the selection of source modules is done on a sample-wise basis. However, these sample-wise approaches do not fully exploit layer-wise and token-wise features.

Therefore, we propose the sequence-level Aligned-Attention, which partially selects the relavant tokens from different source module, and aligns them in the combined sequence. This layer aligns the output tokens of the QKV-MLP and FFN modules at the sequence level:

$$\text{Aligned-Attention}(\mathbf{q}, K, V) = \text{Softmax}\left(\frac{\mathbf{q}K^\top}{\sqrt{d_k}}\right)V \tag{7}$$

where $\mathbf{q} \in \mathbb{R}^{1 \times d_k}$ and $K, V \in \mathbb{R}^{T \times d_k}$. $\mathbf{q}$ represents the query vector, while K and V denote the key and value matrices, respectively. The K and V are obtained by concatenating all the output tokens processed by different source modules in

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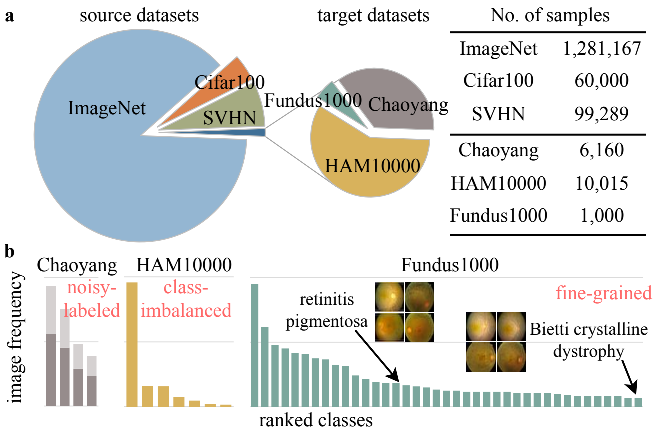

Fig. 3. Datasets statistics. **a**. The target medical datasets are all small-scale compared to the source datasets. **b**. The sample distributions of the ranked classes in the target datasets. The target tasks represent three common medical tasks: noisy-labeled (noisy samples in light gray), class-imbalanced, and fine-grained tasks.

the original sequence order. If we directly extend the routing machanism in vanilla MoE to attention operation, it may block the knowledge flow among different source modules. Therefore, the proposed Aligned-Attention maintains token-level routing and enables sequence-level communication in transformers.

C. Method Details

In our experiments, we follows [42] and employ the ViT-s model for both the source and target tasks. For the input embedding layer, we use single token-level Routing and Fusion layers to process the input patches. To prevent training collapse to local minima when stacking multiple routing modules, we employ several techniques. Firstly, an auxiliary load balancing loss is incorporated for the Routing module, following the vanilla MoE approach, to equalize the allocation of tokens to different source models. Additionally, an early stopping strategy is adopted, where we initially train r models for $1/r$ of the total epochs, and subsequently continue training the agent with the best performance on the validation set. Lastly, every s Mixing Attention layers share the same Routing module, reducing the training complexity. In our experiments, r and s are set to 20 and 3, respectively.

IV. Experiments

In this section, we first introduce the datasets and experimental settings used in our main experiments, which involve three source datasets and three target datasets. Next, we present and analyze the detailed results of our MAT method alongside important baseline methods on the three primary target datasets. We then conduct comprehensive ablation studies to examine each component of our proposed approach. Finally, we provide a range of additional experiments and analyses to further evaluate the generalization capability and computational complexity of our method.

A. Datasets and Experimental Settings

The statistics of the source and target datasets are shown in Figure 3. We utilized ImageNet-1k [43], Cifar100 [44], and SVHN [45] as the source datasets as they are easy to access and have widespread usage [46]. We conducted experiments on three target medical datasets: Chaoyang [47], HAM10000 [48], and Fundus1000 [49]. These datasets, apart from their small-scale nature, exhibit characteristics commonly encountered in real medical applications, such as noisy labels, class imbalance, and fine-grained categories.

In all of the source and target datasets, we resized the images to 224x224. During the training of the source datasets, we performed a set of random image augmentations, including shearing, translation, rotation, brightness, contrast, sharpness adjustment, color jittering, posterizing, solarizing and inverting. We also applied a MixUp [50] training strategy. This variety of augmentations is used to improve the generalization ability of the source models. For the training of the target medical datasets, we applied random horizontal flipping, shearing, and translation for the best downstream performance.

All experiments were conducted on NVIDIA 3090 GPUs. The training of the source tasks used AdamW optimizer with a cosine warmup scheduler. The learning rate was set to 0.001 and the batch size was set to 256. A cosine annealing learning rate scheduler was also used. The training of the target tasks used SGD optimizer with momentum. The learning rate was set to 0.01 and the batch size was set to 64.

To assess the performance of MAT, following [14], [22], we first compared it with the model trained from scratch and models transferred from single-sources. Additionally, comparisons were also made with some SOTA multi-source transfer learning methods, including the ensemble model [11], Zoo-Tuning [22], Hub-Pathway [14], and the modified versions of the vanilla MoE [29]. Note that these methods are not specifically designed for transformer models. Since Zoo-Tuning was originally designed for convolutional models, we evaluated its performance on both ViT-s and ResNet-50, referred to as Zoo-Tuning and Zoo-Tuning$_{\text{ResNet}}$, respectively. The modified variants of MoE are denoted as MoE$_{\text{ImageNet}}$ and MoE$_{\text{random}}$, based on the ways for selecting the QKV-MLP layer (see the "Ablation Study For the Mixing Attention Layer" section for details). We employed some widely-used classification metrics [51], encompassing AUC, AP, precision, Jaccard index, F1-score, micro accuracy, and macro accuracy. Except for the micro accuracy, all metrics were calculated on a per-class basis, which can provide a more comprehensive evaluation of the model performance for imbalanced classes. Each model was trained five times with different seeds. For better comparison, the mean and standard deviation for all the metrics were reported in percentage value.

B. Major Results on Small-scale Medical Datasets

1) Noisy-Labeled Task on Chaoyang Dataset: First, we evaluated MAT on the small-scale Chaoyang dataset, which is a histopathology colorectal cancer classification dataset with four classes: normal, serrated, adenocarcinoma, and adenoma. The dataset consists of colon tissue images collected from

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TABLE I
RESULTS ON THE NOISY-LABELED CHAOYANG DATASET

Model	AUC	AP	precision	Jaccard	F1	micro acc.	macro acc.
Scratch	91.17±0.28	75.30±0.67	71.08±0.78	55.53±0.92	69.31±0.84	76.70±0.74	68.35±0.96
ImageNet*	94.25±0.16	82.41±0.54	72.71±0.97	63.55±0.75	76.12±0.68	81.89±0.36	75.68±0.62
Cifar100*	92.45±0.60	77.61±1.10	72.72±1.05	58.35±0.97	71.92±0.86	78.63±0.78	71.41±0.84
SVHN*	91.41±0.19	74.85±0.46	69.95±0.27	55.41±0.32	69.28±0.37	76.44±0.24	68.84±0.64
Ensemble	94.79±0.08	**84.48**±0.27	78.89±0.85	64.41±1.09	76.99±0.85	82.75±0.63	75.77±0.81
Zoo-Tuning	59.11±8.16	33.30±8.36	25.92±18.56	14.97±9.36	22.57±13.86	37.57±13.28	31.23±7.83
Zoo-Tuning$_{\text{ResNet}}$	92.05±2.56	76.89±4.48	72.40±3.64	58.00±4.93	71.22±4.55	78.60±3.31	70.64±4.66
Hub-Pathway	94.25±0.18	83.14±0.29	78.10±0.36	64.71±0.53	77.14±0.29	82.58±0.49	76.49±0.32
MoE$_{\text{ImageNet}}$	94.52±0.46	83.16±1.07	77.90±1.10	64.90±1.54	77.23±1.25	82.73±0.92	76.83±1.22
MoE$_{\text{random}}$	94.84±0.54	83.80±1.60	78.09±1.39	65.40±1.57	77.60±1.27	83.17±1.21	77.24±1.18
MAT (ours)	**94.89**±0.36	84.07±0.34	**79.12**±0.47	**66.38**±0.34	**78.43**±0.19	**83.85**±0.30	**77.91**±0.34

*: single-source transfer. acc.: accuracy. bold: best result. underlined: second best result.

Beijing Chaoyang Hospital, Capital Medical University, and scanned using a 20x objective lens. It comprises 4,021 histopathology images, with approximately 40% noisy samples in its training set. In professional medical tasks, even involving multiple experienced doctors or radiologists for annotation, label inconsistencies may still persist, resulting in noisy labels. Therefore, the ability to learn from noisy samples is important for deep-learning models in medical tasks.

Table I presents the performance comparison of MAT with other transfer learning approaches. The ensemble method was slightly better (0.86 and 0.09 in micro and macro accuracy respectively) than the best single-source transfer model. Zoo-Tuning with ViT performed poorly due to the heavy Channel Alignment when applied to transformer layers. Moreover, Zoo-Tuning$_{\text{ResNet}}$ also failed to outperform the best single-source transfer model. Hub-Pathway surpassed the best single-source transfer model by utilizing a Pathway Route to multiple source models. However, this routing occurs at the model level and may not fully leverage the rich intermediate features within the models. In contrast, our token- and layer-level MAT outperformed Hub-Pathway by 1.27 and 1.42 in micro and macro accuracy, respectively. This demonstrates that MAT successfully merges the knowledge from multiple sourcs at different layers, resulting in improved accuracy. MoE$_{\text{ImageNet}}$ and MoE$_{\text{random}}$ also surpassed Hub-Pathway and Zoo-Tuning. This further highlights the importance of intermediate feature fusion and conditional activation for training robust models. However, their behavior requires that all the single-source models behaves similarly. When some of the source models cannot perform well, the performance of MoE variants may be affected, which will be shown in the following two datasets and discussed in the ablation study section. We also observed that, compared to the following two datasets, single-source knowledge transfer is difficult on this dataset, especially from SVHN. However, thanks to the integration of token-level Routing and sequence-level Aligned-Attention, MAT can still effectively leverage the knowledge from all source models, resulting in the best performance.

2) Class-Imbalanced Task on HAM10000 Dataset: Here we evaluated MAT on the small-scale HAM10000 dataset, which is a class-imbalanced skin lesion dataset containing 10,015 dermatoscopic images. The collected dermatoscopic images were from different populations, acquired and stored by different modalities. The dataset comprises a representative set of all major diagnostic categories of pigmented skin lesions. Specifically, the images are categorized into seven classes: akiec, bcc, bkl, df, mel, nv, and vasc. Notably, the category with the most samples (nv) has approximately 58 times as many samples as the category with the fewest (df). In real clinical scenarios, imbalances in the dataset are frequently attributed to the varying prevalence of different diseases. Consequently, the performance of models for class-imbalanced tasks holds significant importance in medical applications.

As shown in Table II, MAT achieved the highest performance across all evaluation metrics. Zoo-Tuning and Zoo-Tuning$_{\text{ResNet}}$ exhibited poor performance as they tended to classify samples as the class with the most samples. Hub-Pathway performed slightly better than the transfer model from ImageNet. The MoE variants were unable to surpass the performance of the best single-source transfer model. This can be attributed to the significant performance gap among the three single-source transfer models. In contrast, our MAT leverages Aligned-Attention to ensure that token-level Routing selects the most suitable source modules from a sequence-level perspective, effectively avoiding routing fallacies into local optima. As a result, MAT outperformed Hub-Pathway by a large margin of 7.08 in macro accuracy and 1.57 in micro accuracy. The significant improvement in macro accuracy on this imbalanced dataset further demonstrates the superiority of our MAT approach.

We further show class-wise accuracy on HAM10000 dataset in Figure 4 and Table III. MAT performs better than other methods on the minority-class (see class df). It also performs well on most of the classes. The ensemble method encountered severe stability issues on some classes, making the average accuracy not even surpassing the best single-source transfer model. This indicates that MAT is more suitable for medical problems with class-imbalance.

3) Fine-Grained Task on Fundus1000 Dataset: Finally, we evaluated MAT on the small-scale Fundus1000 dataset with 1,000 images. This dataset consists of 39 fine-grained fundus diseases and conditions. All the fundus images were collected from the Joint Shantou International Eye Centre (JSIEC) in Shantou, Guangdong Province, China. In medical image analy-

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TABLE II
RESULTS ON THE CLASS-IMBALANCED HAM10000 DATASET

Model	AUC	AP	precision	Jaccard	F1	micro acc.	macro acc.
Scratch	$93.47_{\pm 0.40}$	$60.29_{\pm 1.44}$	$66.70_{\pm 3.96}$	$41.88_{\pm 1.26}$	$54.72_{\pm 1.13}$	$77.50_{\pm 1.13}$	$53.41_{\pm 1.68}$
ImageNet*	$97.96_{\pm 0.25}$	$84.55_{\pm 1.85}$	$79.26_{\pm 1.34}$	$64.87_{\pm 1.09}$	$77.83_{\pm 0.74}$	$88.44_{\pm 0.49}$	$\underline{77.24}_{\pm 2.63}$
Cifar100*	$96.37_{\pm 0.35}$	$77.64_{\pm 1.49}$	$74.36_{\pm 1.68}$	$58.25_{\pm 1.32}$	$72.02_{\pm 1.02}$	$83.82_{\pm 0.62}$	$70.73_{\pm 2.02}$
SVHN*	$94.01_{\pm 0.34}$	$63.88_{\pm 0.87}$	$66.56_{\pm 4.25}$	$46.99_{\pm 2.81}$	$60.15_{\pm 3.07}$	$79.36_{\pm 0.99}$	$58.41_{\pm 2.89}$
Ensemble	$\underline{98.15}_{\pm 0.12}$	$\underline{87.06}_{\pm 1.13}$	$83.66_{\pm 1.38}$	$\underline{67.97}_{\pm 1.28}$	$\underline{79.76}_{\pm 1.00}$	$88.35_{\pm 0.55}$	$77.07_{\pm 1.12}$
Zoo-Tuning	$50.48_{\pm 3.13}$	$14.82_{\pm 0.76}$	$9.53_{\pm 0.00}$	$9.53_{\pm 0.00}$	$11.44_{\pm 0.00}$	$66.73_{\pm 0.00}$	$14.29_{\pm 0.00}$
Zoo-Tuning$_{\text{ResNet}}$	$87.50_{\pm 4.53}$	$38.85_{\pm 9.45}$	$33.53_{\pm 9.76}$	$20.44_{\pm 6.45}$	$27.64_{\pm 9.24}$	$71.55_{\pm 3.66}$	$28.51_{\pm 8.50}$
Hub-Pathway	$97.95_{\pm 0.07}$	$86.51_{\pm 1.21}$	$\underline{83.68}_{\pm 1.92}$	$66.72_{\pm 1.41}$	$79.11_{\pm 0.81}$	$\underline{88.58}_{\pm 0.64}$	$76.00_{\pm 0.59}$
MoE$_{\text{ImageNet}}$	$97.33_{\pm 1.44}$	$80.10_{\pm 8.50}$	$76.82_{\pm 8.97}$	$61.35_{\pm 8.42}$	$73.75_{\pm 8.23}$	$86.57_{\pm 3.42}$	$72.26_{\pm 8.38}$
MoE$_{\text{random}}$	$94.23_{\pm 1.22}$	$64.52_{\pm 5.77}$	$62.52_{\pm 7.29}$	$45.70_{\pm 5.59}$	$58.69_{\pm 5.85}$	$79.32_{\pm 2.21}$	$57.82_{\pm 5.92}$
MAT (ours)	$\mathbf{98.45}_{\pm 0.18}$	$\mathbf{90.85}_{\pm 0.44}$	$\mathbf{84.66}_{\pm 1.89}$	$\mathbf{72.52}_{\pm 1.06}$	$\mathbf{83.56}_{\pm 0.67}$	$\mathbf{90.15}_{\pm 0.42}$	$\mathbf{83.08}_{\pm 1.15}$

*: single-source transfer. acc.: accuracy. bold: best result. underlined: second best result.

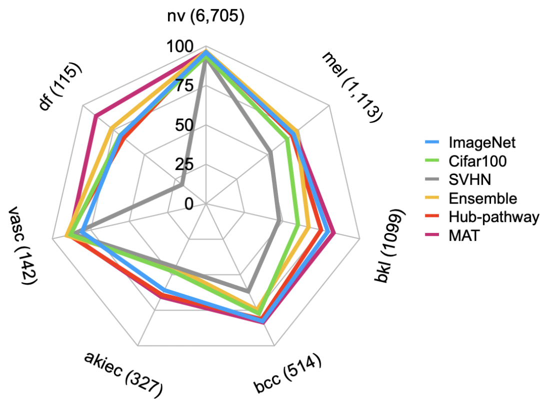

Fig. 4. Per-class accuracy on HAM10000 dataset with 7 classes. Number of samples for each class is shown in brackets.

TABLE III
PER-CLASS ACCURACY ON HAM10000 DATASET

class	nv	mel	bkl	bcc	akiec	vasc	df	avg.
ImageNet	96.05	70.95	<u>79.32</u>	<u>82.71</u>	60.87	80.72	69.33	<u>77.24</u>
Cifar100	93.54	65.79	59.86	77.36	47.81	88.13	69.62	70.73
SVHN	93.01	52.34	47.60	61.76	44.64	86.39	19.45	58.41
Ensemble	**96.63**	**73.52**	66.75	74.68	45.01	**93.57**	<u>76.64</u>	77.07
Hub-Pathway	96.34	69.82	74.88	81.25	<u>64.25</u>	<u>89.20</u>	66.74	76.00
MAT	<u>96.44</u>	<u>72.21</u>	**83.04**	**83.57**	**65.56**	88.40	**89.21**	**83.08**

sis, diseases with similar visual characteristics may necessitate varying treatment and prognosis. Therefore, the ability of deep learning models to perform fine-grained classification is crucial in medical tasks.

As Table IV shows, our MAT achieved the best average performance. Except for the micro accuracy, ensemble method is slightly better than the best single-source model. Both Zoo-Tuning and Zoo-Tuning$_{\text{ResNet}}$ performed poorly. While Hub-Pathway still outperformed the best single-source transfer model, the improvement was marginal. MoE$_{\text{ImageNet}}$ performed slightly worse than the transfer model from ImageNet, likely due to the negative influence of the poorly performing models from Cifar100 and SVHN. MoE$_{\text{random}}$ performed even worse than MoE$_{\text{ImageNet}}$ because it has more parameters from Cifar100 and SVHN. The large standard deviation of MoE$_{\text{random}}$ also indicates that the random choice of differently-performing source models negatively affects its performance. Nonetheless, our MAT still outperformed Hub-Pathway by 4.54 and 2.15 on macro and micro accuracy, respectively, demonstrating its consistent capability in fine-grained classification tasks.

C. Ablation Study and Hyper-parameter Analysis

1) Ablation Study for the Mixing Attention Layer: MAT is designed to extract and leverage knowledge from multiple source models for transformer models. However, the self-attention layer in the original transformers cannot be routed token-independently as it requires sequence-level attention operations. As a result, the vanilla MoE only splits the FFN layer into multiple experts. While this is sufficient for learning from scratch, it cannot fully utilize all the source parameters in multi-source transfer learning. To address this, we split the original self-attention layer into two parts: the QKV-MLP and Attention layers. In the Mixing Attention Layer, the QKV-MLP layers are routed to multiple sources, while the Attention layers are merged into a single sequence-level Aligned-Attention layer.

We conducted several ablation studies for the Mixing Attention layer. First, we compared MAT with two MoE variations: MoE$_{\text{ImageNet}}$ and MoE$_{\text{random}}$. MoE$_{\text{ImageNet}}$ initializes the QKV-MLP parameters from the source model from ImageNet. MoE$_{\text{random}}$ randomly selects one from the source models for each QKV-MLP. In addition, we compared MAT to the versions without Token-Routing and Aligned-Attention modules, namely MAT$_{\text{w/o Token-Routing}}$ and MAT$_{\text{w/o Aligned-Attention}}$. In the absence of Token-Routing, MAT allocates input tokens to all source modules with equal weights. In the absence of Aligned-Attention, each source QKV-MLP has its own attention operation.

The ablation studies of MAT$_{\text{w/o Token-Routing}}$ and MAT$_{\text{w/o Aligned-Attention}}$ are shown in Table V. The results of vanilla MoE variations are shown in Table I, II, and IV. When all the single-source transfer models performed well, like on the Chaoyang dataset, the MoE variations outperformed the best single-source model. This indicates that fusing multi-source knowledge helps improve performance. However, they are more susceptible to the impact of poorly-performed source models, as observed on the HAM10000 and Fundus1000 datasets. This phenomenon also occurred to MAT$_{\text{w/o Aligned-Attention}}$, which performed even worse,

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TABLE IV
RESULTS ON THE FINE-GRAINED FUNDUS1000 DATASET

Model	AUC	AP	precision	Jaccard	F1	micro acc.	macro acc.
Scratch	$91.45_{\pm 1.28}$	$55.86_{\pm 1.45}$	$38.65_{\pm 1.50}$	$32.67_{\pm 1.58}$	$39.71_{\pm 1.77}$	$51.87_{\pm 1.36}$	$45.40_{\pm 3.02}$
ImageNet*	$99.73_{\pm 0.12}$	$93.29_{\pm 2.65}$	$78.22_{\pm 3.80}$	$73.28_{\pm 2.80}$	$78.27_{\pm 3.11}$	$85.94_{\pm 2.17}$	$81.59_{\pm 3.07}$
Cifar100*	$98.84_{\pm 0.18}$	$82.11_{\pm 2.41}$	$61.57_{\pm 3.99}$	$54.57_{\pm 4.96}$	$62.22_{\pm 4.31}$	$73.12_{\pm 3.24}$	$68.23_{\pm 4.30}$
SVHN*	$94.91_{\pm 0.76}$	$68.31_{\pm 1.86}$	$50.38_{\pm 4.33}$	$41.53_{\pm 4.42}$	$49.97_{\pm 4.36}$	$59.58_{\pm 2.89}$	$54.83_{\pm 3.88}$
Ensemble	$\underline{99.74}_{\pm 0.05}$	$\underline{93.64}_{\pm 2.71}$	$77.23_{\pm 3.50}$	$75.30_{\pm 3.31}$	$78.45_{\pm 3.02}$	$\mathbf{89.93}_{\pm 0.60}$	$81.60_{\pm 2.58}$
Zoo-Tuning	$63.11_{\pm 3.64}$	$11.02_{\pm 1.57}$	$1.87_{\pm 0.89}$	$1.84_{\pm 0.89}$	$2.68_{\pm 1.10}$	$19.58_{\pm 3.41}$	$6.01_{\pm 1.43}$
Zoo-Tuning$_{\text{ResNet}}$	$90.97_{\pm 1.70}$	$52.45_{\pm 6.67}$	$34.20_{\pm 8.86}$	$27.57_{\pm 7.09}$	$34.01_{\pm 7.81}$	$48.33_{\pm 6.57}$	$37.80_{\pm 7.45}$
Hub-Pathway	$99.21_{\pm 0.16}$	$90.11_{\pm 0.98}$	$\underline{79.28}_{\pm 1.37}$	$\underline{75.51}_{\pm 1.60}$	$\underline{79.74}_{\pm 1.41}$	$86.81_{\pm 0.60}$	$\underline{82.56}_{\pm 1.31}$
MoE$_{\text{ImageNet}}$	$99.65_{\pm 0.16}$	$93.26_{\pm 2.99}$	$75.97_{\pm 1.61}$	$72.13_{\pm 2.07}$	$76.68_{\pm 1.58}$	$85.00_{\pm 1.40}$	$79.81_{\pm 1.42}$
MoE$_{\text{random}}$	$95.60_{\pm 1.63}$	$65.24_{\pm 9.60}$	$47.40_{\pm 8.45}$	$40.30_{\pm 9.18}$	$47.32_{\pm 8.96}$	$59.79_{\pm 9.01}$	$52.01_{\pm 9.18}$
MAT (ours)	$\mathbf{99.76}_{\pm 0.05}$	$\mathbf{95.00}_{\pm 1.03}$	$\mathbf{86.74}_{\pm 2.47}$	$\mathbf{81.65}_{\pm 2.78}$	$\mathbf{85.54}_{\pm 2.55}$	$\underline{88.96}_{\pm 1.19}$	$\mathbf{87.10}_{\pm 2.62}$

*: single-source transfer. acc.: accuracy. bold: best result. underlined: second best result.

TABLE V
ABLATION STUDY. WE PRESENT THE F1 SCORES FOR EACH DATASET

Model	Chaoyang	HAM10000	Fundus1000
Diff.$_{\text{ImageNet-SVHN}}$*	6.84	16.78	28.3
MAT$_{\text{w/o Token-Routing}}$	71.24	59.24	44.54
MAT$_{\text{w/o Aligned-Attention}}$	73.48	64.76	49.43
MAT	**78.43**	**83.56**	**85.54**

*: performance gap between the best and worst single-source transfer models from ImageNet and SVHN.

indicating that without the Aligned-Attention layer, the routing for QKV-MLP is ineffective, even worse than not routing the QKV-MLP. In contrast, our MAT achieved the best performance on all three datasets, even in cases where some of the source models underperformed. MAT effectively leverages valuable knowledge while discarding less informative components. These results demonstrate that MAT successfully extracts pertinent knowledge from multiple source models.

2) Analysis on Token-Routing: The Token-Routing distributions are illustrated in Figure 5. Our observations indicate that the performance of MAT is mainly dependent on the single-source transfer model from ImageNet for the top-1 routing. The top-2 routing is more balanced, with the source models from Cifar100 and SVHN contributing more to the routing. However, we also note that different layers exhibit varying behaviors. The parameters learned from ImageNet are particularly effective in extracting low-level features, while high-level knowledge can be acquired from all the source models. These findings demonstrate that the fusion of knowledge from multiple sources can vary across layers. Consequently, token- and layer-level MAT outperformed model-level fusion methods such as Hub-Pathway.

We also performed a qualitative visualization of the Token-Routing in Figure 6. The image patches routed to different source models via Token-Routing exhibit distinct distributions. Furthermore, the ability of each source model is somewhat related to the source dataset. The modules from Cifar100 mostly capture blurry boundary lines. The modules from SVHN excel at capturing small, detailed dots, lines, and background patches. On the other hand, the ImageNet modules exhibit a more generalist behavior, being routed for patches with a wide range of features.

The effect of the parameter k in the top-k routing strategy is illustrated in Figure 7. When k is set to 1, the performance is notably poor, especially on the Fundus1000 dataset. This can be attributed to the auxiliary load balancing loss, which significantly affects the routing process when k=1. On the other hand, when k is set to 2 or 3, the performance is similar, with k=2 slightly outperforming k=3. Therefore, we set k=2 for our experiments.

We also conducted experiments with different hyper-parameters in Table VI, including the minimal noise ϵ in Equation 3, early stopping ratio r, and routing sharing frequency s in Method Details section. The results demonstrate that ϵ should not be set too large, and that a moderate increase in r and s can further improve the performance of MAT.

TABLE VI
DIFFERENT HYPER-PARAMETERS ON CHAOYANG

	$\epsilon = 0$	$\epsilon = 0.01$	$\epsilon = 0.1$	$\epsilon = 1.0$
macro acc.	77.86	**77.91**	77.73	76.57
	$r = 1$	$r = 10$	$r = 20$	$r = 50$
macro acc.	76.61	77.75	77.91	**78.02**
	$s = 1$	$s = 2$	$s = 3$	$s = 6$
macro acc.	77.05	77.54	**77.91**	77.76

3) Analysis on Data Sampling Ratio: To futher investigate the influence of the scale of target datasets, we conducted experiments with different ratios of training data on all the three datasets. We use 10%, 30%, 50%, and 80% of the training data to train Hub-Pathway and our MAT. The results in Figure 8 indicate that MAT drops a little on performance with fewer data on HAM10000 and Chaoyang datasets. The size of Fundus1000 is too small that the performance of MAT is not stable with low data sampling ratios. Although the advantage of MAT becomes less pronounced when the number of samples is below a few hundred, its performance remains at least comparable to the baselines. In these experiments, MAT outperforms Hub-Pathway and Ensemble methods on all the datasets with different data sampling ratios.

D. Additional Experiments

1) Analysis of More Datasets and Tasks: We also conducted experiments with two more medical sources to view the

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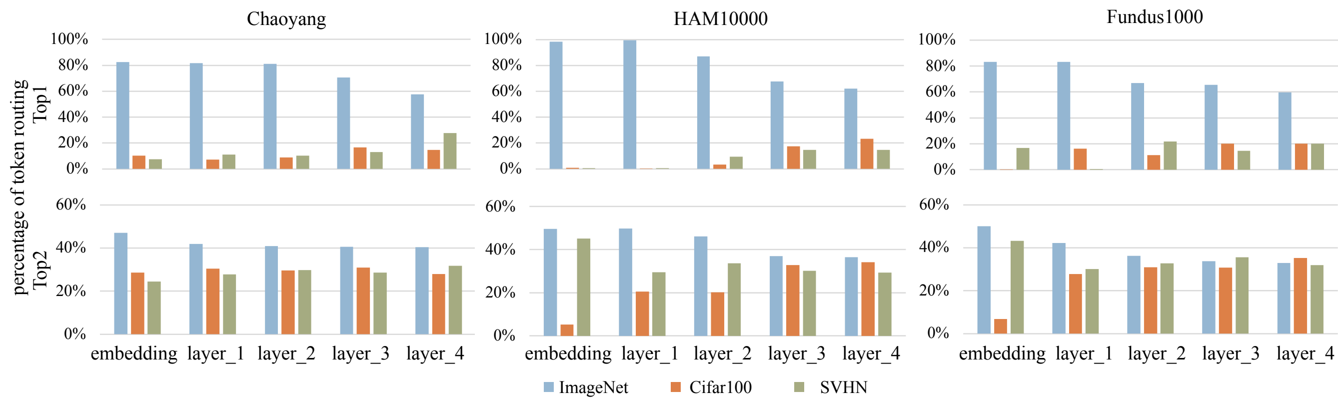

Fig. 5. Top-1 and top-2 Token-Routing analysis. We summarize the number of input tokens routed to each source-module in each layer.

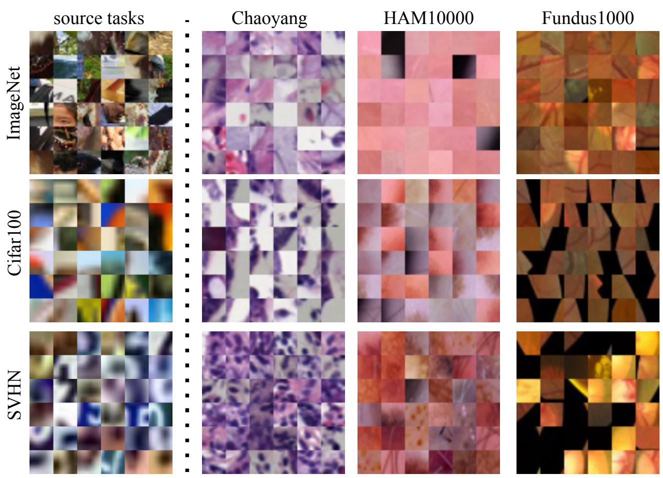

Fig. 6. Qualitative Visualization of Token-Routing. Image patches are randomly sampled from each dataset. Patches in the target datasets are grouped according to the top-1 gating weights of the Token-Routing in the first Mixing Attention layer. Different source modules exhibit varying capabilities in extracting image features, which are also somewhat related to the source dataset.

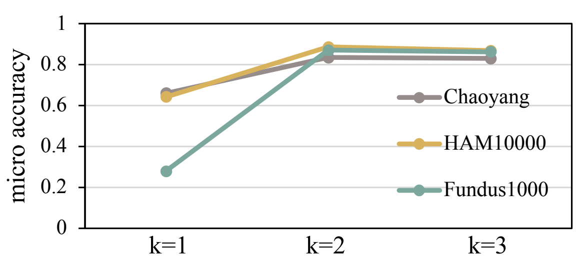

Fig. 7. The choice of top-k strategy in the Token Routing module.

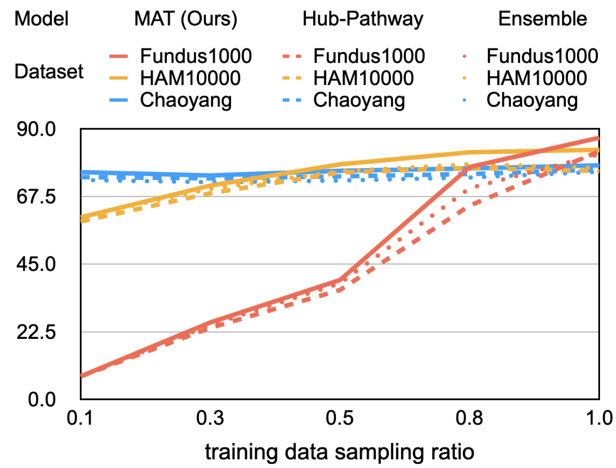

Fig. 8. Macro accuracy on the test sets of different training data sampling ratios

performance of MAT on more sources. We added two medical source datasets: IU-Xray frontal and lateral [52]. They contain the frontal and lateral x-ray images. In addition to the three aforementioned sources, there are five source datasets included here. Table VII shows that MAT still achieves the best performance. Besides, it is better than MAT transfered from 3 sources, indicating the effectiveness of MAT to merge knowledge from more sources. Due to the typically limited scale and varied features of medical datasets, direct single-source transfer can result in poor performance (like that from IU-Xray frontal). Thus, it becomes crucial to extract pertinent information while filtering out irrelevant factors for the target tasks. Our MAT is designed to address this challenge by integrating insights from diverse sources.

As for how to select the hyper-param k that merges the best top-k sources for knowledge fusion in MAT, we consider the best balance between performance and complexity. Therefore, we still use k=2 in this 5 sources settings (see Table VIII).

In addition, we conducted a multi-task pre-training experiment called "joint" in Table VII. In this setting, all source datasets are gathered and pretrained simultaneously. Since the multi-task learning is also challenging task due to the negative impact of task dissimilarities, the joint method does not improve the transfer noticeably as shown in Table VII. Moreover, sources may not be available simultaneously in real-world applications, especially for medical tasks, which motivates us to focus on the multi-source transfer learning.

Moreover, to see how MAT performs on other tasks, we conducted an experiment on a 3D dataset - COVID-CT [53], and a segmentation dataset - ORIGA-light [54]. COVID-CT is a CT image classification dataset, and the ORIGA-light is a fundus segmentation dataset. We evaluated the accuracy and Dice score on these two datasets, using three source datasets to transfer. In Table IX, MAT still achieves the SOTA performance.

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TABLE VII
MACRO ACC. WITH TWO MORE SOURCE TASKS

Dataset	single source					3 sources		5 sources		
	ImageNet	Cifar100	SVHN	**IU-Xray$_{\mathbf{frontal}}$**	**IU-Xray$_{\mathbf{lateral}}$**	Joint	MAT	Ensemble	Hub-Pathway	**MAT**
Chaoyang	75.68	71.41	68.22	64.76	69.68	75.79	77.91	75.10	76.32	**78.51**
HAM10000	77.24	70.73	58.41	52.72	55.19	75.84	83.08	75.35	76.62	**83.57**
Fundus1000	81.59	68.23	54.83	49.13	51.68	81.10	87.10	80.12	82.13	**87.18**

TABLE VIII
MACRO-ACCURACY OF DIFFERENT TOP-K TRANSFERRED FROM TWO ADDITIONAL SOURCES

Dataset	Top-1	Top-2	Top-3	Top-4	Top-5
Chaoyang	53.66	78.51	77.89	78.53	**78.75**
HAM10000	60.12	83.57	83.61	83.49	**83.64**
Fundus1000	63.73	87.18	**87.22**	86.94	87.10

TABLE IX
ACC AND DICE ON ADDITIONAL 3D AND SEGMENTATION DATASETS

Dataset	ImageNet	Cifar100	SVHN	Ensemble	Hub-Pathway	MAT
COVID-CT	84.02	82.95	80.33	83.94	84.31	**84.88**
ORIGA-light	86.85	86.54	86.19	87.17	86.72	**87.53**

2) Different backbones and more compatible method: Here, to see how Mixing Attention Transfer learning method performs on different backbones, we replaced the ViT-s backbone to Swin-s [55]. Swin Transformer is also transformer-based model, with more complicate architecture than ViT. But it introduces more inductive biases to the model, which is helpful in small-scale medical datasets. The results are shown in Table X. The performance of MAT with Swin-s backbone is slightly better than that with ViT-s backbone, except on the HAM10000 dataset. Our MAT method still outperforms the other methods, which indicates the effectiveness of our method on various backbones.

TABLE X
MACRO ACCURACY OF FST-TSC AND SWIN-S

Dataset	ViT-s		Swin-s		
	FST-TSC	MAT	Ensemble	Hub-Pathway	MAT
Chaoyang	28.67	77.91	78.59	78.67	**80.02**
HAM10000	35.32	**83.08**	80.34	79.01	82.36
Fundus1000	40.29	87.10	82.47	84.58	**88.39**

Transfer learning from multiple sources is a challenging task. Few methods can effectively utilize knowledge from multiple sources without being negatively impacted by task dissimilarities. Here, we analized another relavant method that can be adopted in multi-source transfer learning, FST-TSC [56]. However, it is not perfectly relevant since it is designed for time series data, which may lead to poor performance on our image classification tasks. We flatten the input image pixels as the sample feature and use FST-TSC to transfer the source data style to the target. Moreover, FST-TSC needs both the source and target datasets available in the fine-tuning stage, which may be impossible in real-world applications. The results are shown in Table X. We can see that FST-TSC performs far poorer on our task, indicating that multi-source transfer learning on image classification tasks is challenging.

3) Complexity Analysis: We present an analysis of the resource and computation complexity in Table XI. The number of parameters and floating point operations (FLOPs) in most multi-source transfer methods is approximately n-times that of the source model, where n is the number of source models. In Zoo-Tuning, additional parameters are introduced for channel alignment weights. This increase the model size particularly for transformer models, since the channel alignment is heavy for fully-connected layers in transformer models. Vanilla MoE only employs multiple source models for the FFN layers, resulting in a slight reduction in parameter and computation complexity. However, this approach also leads to a significant drop in performance (6.6 and 21.28 for MoE$_{\text{ImageNet}}$ and MoE$_{\text{random}}$ respectively). Because of the top-k routing strategy adopted by MAT, only k/n source modules are activated for each input, thereby reducing the computation complexity to k/n compared to typical multi-source transfer methods.

TABLE XI
PERFORMANCE AND COMPLEXITY ANALYSIS. THE F1 SCORES ARE AVERAGED ON CHAOYANG, HAM10000, AND FUNDUS1000 DATASETS

Model	Params (M)	FLOPs (G)	avg. F1
Scratch	21.59	4.58	54.58
ImageNet	21.59	4.58	77.41
Cifar100	21.59	4.58	68.72
SVHN	21.59	4.58	59.80
Ensemble	64.76	13.75	78.40
Zoo-Tuning	209.11	66.14	12.23
Zoo-Tuning$_{\text{ResNet}}$	79.05	17.79	44.29
Hub-Pathway	69.67	10.06	78.66
MoE$_{\text{ImageNet}}$	49.99	7.37	75.89
MoE$_{\text{random}}$	49.99	7.37	61.21
MAT	64.81	9.01	82.51

Although MAT requires more parameters than the single-source transfer models, the performance improvement is significant. Since medical datasets are often small-scale, well-initialized model weights can help reduce overfitting. The additional parameters in MAT are initialized by multiple source models, they merge and transfer knowledge from multiple sources to increase the F1 score by 5.08 on average compared to the best single source transfer model from ImageNet.

V. CONCLUSION

In this study, we propose Mixing Attention Transfer (MAT), a novel multi-source transfer learning approach of transformers for medical image analysis. MAT enhances the performance of the target model by incorporating knowledge from multiple source tasks via the Mixing Attention layer. We conducted evaluations of MAT on three medical datasets, where MAT outperforms state-of-the-art methods due to its effective fusion

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of source knowledge. It also suggested that effectively transferring from multiple sources is better than single source for small-scale medical tasks. MAT has a limitation that it is only designed for vanilla transformers. However, since the transformer model's widespread applicability and strong performance in many tasks, MAT is still a technically-promising multi-source transfer learning approach. For example, we achieved state-of-the-art results on the Chaoyang dataset [57]. Our experiments highlight the robust performance of MAT in tackling prevalent challenges in medical settings, including noisy-labeled, class-imbalanced, and fine-grained tasks.

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